

$$f(x) = \frac{g(x)}{\int g(x)dx}.$$

Using the $\propto$ relation, show that

$$p_{pos}(\mu | D) \propto \mu^{a+m-1} (1-\mu)^{b+n-m-1}.$$

Conclude then that the posterior distribution for μ must be a Beta distribution of the form $Beta(a+m, b+n-m)$.

Problem 2 (Ridge Regression as Maximum a Posteriori Estimation)

Recall the ridge estimator:

$$\tilde{\beta}_R = \arg \min_{\beta} \sum_{i=1}^n (y_i - x_i^\top \beta)^2 + \alpha \sum_{i=1}^p \beta_i^2.$$

The Bayesian maximum-a-posteriori (MAP) estimator for linear regression is:

$$\tilde{\beta}_{MAP} = \arg \max_{\beta} p(\beta | Z),$$

where $p(\beta | Z)$ is the pdf of posterior distribution resulting from Bayesian inference on the data $Z = \{(x_1, y_1), \dots, (x_n, y_n)\}$. This pdf is given by

$$p(\beta | Z) = \frac{p(Z | \beta)p(\beta)}{p(Z)} \propto p(Z|\beta) p(\beta) \quad (5)$$

We assume that the prior for β is $\mathcal{N}(0, s_0^2 I)$ and that the likelihood $p(Z | \beta)$ is given by,

$$p(Z | \beta) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2} (y_i - x_i^\top \beta)^2\right). \quad (6)$$

Show that $\tilde{\beta}_R = \tilde{\beta}_{MAP}$, when $\lambda = \sigma^2/s_0^2$.

Hint: $\arg \max_{\beta} f(\beta) = \arg \max_{\beta} \log f(\beta)$. Note also that $p(Z)$ does not depend on β !

Hint:
remember $p(Z)$ is like a const. when differentiating w.r.t. β

① $\propto p(Z|\beta) p(\beta)$

② plug in